INDEPENDENT RESEARCH PROJECT

Predictive Modelling and Optimisation of Microwave Plasma CVD Diamond Growth Using Experimental Data

Materials Engineering · Computational Modelling

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Independent research and software development portfolio project

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Abstract

Microwave plasma chemical vapour deposition (MPCVD) produces diamond materials for electronics, thermal management, optical systems and advanced manufacturing (Butler et al., 2009). This project examines how exploratory computational modelling can analyse relationships between selected operating conditions and diamond growth.

A Python framework was developed using six observations reported by Martyanov et al. (2023). Multiple linear regression models growth rate from methane concentration and substrate temperature, while quadratic regression explores the nonlinear relationship between methane concentration and growth rate. The framework includes data processing, scientific visualisation, scenario analysis, an illustrative thickness calculation and a Streamlit interface.

The models were evaluated using in-sample metrics and leave-one-out cross-validation (LOOCV). Although the quadratic model performed better than the linear model, both models produced negative cross-validated R^2 values, indicating poor held-out predictive performance. The reference calculation predicts a growth rate of 1.762 µm/h and an estimated thickness of 8.810 µm after five hours.

The dataset does not fully represent the original reactor conditions, including pressure and microwave power, and the models have not undergone independent experimental validation. Their outputs are therefore exploratory and should not be used as reliable industrial predictions.

Overall, the project demonstrates a reproducible integration of materials-science research, regression modelling and software engineering while identifying the need for larger datasets, uncertainty analysis, cross-validation and independent experimental testing.

Chapter 1: Introduction

1.1 Background

Diamonds are technologically significant because of their outstanding mechanical, thermal, and electrical properties. While natural diamonds have been used primarily for jewellery and cutting tools, synthetic diamonds have grown increasingly important in advanced materials engineering (Butler et al., 2009).

One well-known technique for synthesising diamonds is chemical vapour deposition (CVD). During CVD, carbon-containing gases are activated in controlled conditions that allow carbon species to participate in diamond growth onto a suitable substrate (Butler et al., 2009).

Microwave plasma chemical vapour deposition (MPCVD) uses microwaves to sustain plasma, usually in a hydrogen gas mixture enriched with methane. The plasma provides reactive species that promote diamond growth, while substrate and reactor parameters affect the deposition process (Butler et al., 2009; Martyanov et al., 2023).

Controlling diamond growth requires consideration of more than deposition rate. Crystal quality, defect density, surface morphology and other material characteristics also determine suitability for a given application (Li et al., 2006; Teraji, 2006).

1.2 Engineering Significance

Diamond's high thermal conductivity, wide energy gap, optical transparency and wear resistance support applications in thermal management, high-power electronics, optical systems and advanced manufacturing (Butler et al., 2009).

Nevertheless, producing diamond with desired characteristics involves challenges in controlling the growth environment (Teraji, 2006).

Relevant MPCVD parameters include methane concentration, substrate temperature, reactor pressure, microwave power and gas composition. Because these variables interact, changes intended to increase growth rate may also affect material quality (Li et al., 2006; Martyanov et al., 2023).

Understanding these interactions is therefore important for diamond production, and computational modelling is one way to investigate them (Sedov et al., 2023).

1.3 Research Problem

MPCVD diamond growth is governed by complex physical and chemical processes (Butler et al., 2009).

Despite experimental data on growth behaviour, creating reliable models requires appropriate datasets.

Simplified regression models can help explore relationships between process variables and observed growth rates. However, their usefulness depends on the quality, quantity and relevance of the available data.

This project studies the feasibility of developing a computational prototype to explore selected MPCVD growth parameters.

The main challenge is to distinguish between a model that produces mathematically consistent outputs and one that can reliably predict physical behaviour.

1.4 Research Aim

This research project aims to design and critically evaluate an exploratory Python-based framework for examining relationships between selected MPCVD process parameters and diamond growth rate.

1.5 Research Objectives

The objectives are to:

1. Investigate the materials-science principles underlying MPCVD diamond growth.
2. Review relevant experimental and computational research.
3. Identify and document suitable experimental data.
4. Implement multiple linear and quadratic regression models.
5. Develop a Python-based computational system for growth-rate prediction and illustrative thickness estimation.
6. Design and develop an interactive Streamlit app for analysing model results.
7. Include automated testing and reproducibility checks.
8. Analyse scientific limitations and determine future experiment needs.

1.6 Research Question

To what extent can Python-based exploratory regression modelling identify relationships between selected MPCVD process parameters and diamond growth rate?

1.7 Scope of the Study

The current implementation focuses primarily on methane concentration and substrate temperature, and a separate quadratic regression model assesses the relationship between methane concentration and diamond growth rate.

The framework estimates deposited diamond thickness by multiplying the predicted growth rate by the deposition duration, assuming the growth rate remains constant.

The model also supports scenario comparisons across selected input-value ranges.

The existing dataset contains six experimental records from Martyanov et al. (2023), with the values verified against the cited source. However, it does not fully represent the experimental conditions, including reactor pressure and microwave power. The evidence is therefore insufficient to support industrial application or validated reactor optimisation.

Accordingly, the project is limited to exploring computational methods and assessing their potential value for future materials-engineering research.

Chapter 2: Literature Review

2.1 Introduction

MPCVD is a well-established process used to fabricate diamond films in controlled reactor environments. The efficiency of this process depends on complex interactions among gas-phase chemistry, plasma behaviour, the substrate environment, and surface growth mechanisms (Butler et al., 2009; Teraji, 2006).

This review examines the scientific principles governing MPCVD diamond growth, emphasising the influence of key operating variables as identified in the literature, and discusses methodological approaches, including both experimental and computational techniques, used to study and model these relationships. It also highlights how mathematical modelling, such as regression analysis and other quantitative frameworks, has been applied to interpret empirical findings and explore parameter dependencies within the broader context of materials science research.

This study focuses on methane concentration and substrate temperature because they are the explanatory variables used in the present computational prototype.

2.2 Physical and Chemical Principles of MPCVD

MPCVD activates a carbon-containing gas mixture using microwave-generated plasma.

Methane is commonly used as a carbon source, while hydrogen plays an important role in the surface reactions associated with diamond deposition (Butler et al., 2009).

Reactive species generated in the plasma contribute to carbon incorporation at the growing surface. However, diamond deposition does not depend solely on carbon availability (Butler et al., 2009).

The concentrations of reactive species, the substrate temperature and surface reactions influence whether the deposit develops diamond-like properties (Butler et al., 2009; Teraji, 2006).

Butler et al. (2009) describe the interacting gas-phase kinetics, plasma processes and surface reactions involved in diamond CVD.

These coupled processes explain why a simple statistical relationship between operating variables and growth rate cannot represent the full physical system.

2.3 Influence of Methane Concentration

Methane concentration is an important variable in MPCVD because it affects the availability of carbon-containing reactive species (Butler et al., 2009).

Li et al. (2006) investigated how methane concentration and reactor pressure affect diamond-film growth. Within their experimental range, they reported that growth rates increased with methane concentration at fixed pressure.

However, this relationship does not hold across all operating regimes. Martyanov et al. (2023) reported a nonlinear trend across methane concentrations from 4% to 100% at 800 °C.

At a substrate temperature of 800 °C, the growth rate increased from 0.60 µm/h at 4% methane to 3.41 µm/h at 20% methane, then decreased to 0.60 µm/h at 100% methane (Martyanov et al., 2023).

These data show that increasing methane concentration does not necessarily lead to a continuously increasing diamond growth rate.

They also provide a scientific motivation for comparing linear and nonlinear regression approaches.

2.4 Influence of Substrate Temperature

Substrate temperature influences diamond synthesis through its effects on surface chemistry, kinetics, and structure (Butler et al., 2009; Teraji, 2006).

According to Teraji (2006), substrate temperature and methane concentration critically affect the growth rate and crystal quality of homoepitaxial diamond growth.

Martyanov et al. (2023) also reported substantial differences in growth behaviour between their 800 °C and 900 °C experimental series.

However, temperature effects cannot be fully separated from other reactor conditions because the experiments adjusted pressure and microwave power (Martyanov et al., 2023).

Consequently, the present computational model should not be interpreted as identifying an independent causal effect of substrate temperature.

2.5 Other Important Process Variables

Additional parameters beyond methane concentration and substrate temperature influence diamond growth (Butler et al., 2009; Li et al., 2006).

These include:

- Reactor pressure
- Microwave power
- Gas composition
- Substrate material and preparation
- Reactor geometry
- Plasma chemistry
- Deposition duration

Research shows that these factors influence both growth rate and material quality (Butler et al., 2009; Teraji, 2006; Martyanov et al., 2023).

Omitting these variables reduces the reliability and physical interpretability of the regression models, especially because reactor pressure and microwave power differed across the selected experimental series.

2.6 Diamond Growth Rate and Material Characteristics

A higher diamond growth rate does not necessarily indicate better material quality (Angus et al., 1995, pp. 45–50; Li et al., 2006; Teraji, 2006).

Important material characteristics include crystallinity, defect density, surface morphology and the proportion of non-diamond carbon phases (Butler et al., 2009; Teraji, 2006).

Martyanov et al. (2023) described the surface morphology and phases of polycrystalline diamond films using methods such as scanning electron microscopy and Raman spectroscopy.

These findings suggest that changes in methane concentration and substrate temperature may affect both structure and growth rate (Martyanov et al., 2023).

Therefore, future optimisation should consider more factors than just growth rate.

2.7 Mathematical and Computational Modelling

Mathematical modelling can analyse empirical trends and formulate hypotheses for further exploration.

Regression provides a comparatively simple approach for exploring dependencies between process variables and growth rate.

Multiple linear regression represents the response as an additive linear function of the explanatory variables.

Quadratic regression can describe simple curvilinear dependencies and may therefore be used to explore non-monotonic behaviour (Cudeck and du Toit, 2002, pp. 501–519).

However, fitting a mathematical function does not prove its physical correctness.

The applicability of a mathematical model depends on the quality and coverage of the experimental data, the model assumptions and independent validation (Anderson and Papachristodoulou, 2009).

For MPCVD, more advanced numerical models may incorporate plasma chemistry and surface-reaction kinetics and then be evaluated against experimental measurements (Butler et al., 2009).

2.8 Research Gap and Project Contribution

The literature provides extensive knowledge of MPCVD diamond growth and the effects of process parameters (Butler et al., 2009; Li et al., 2006; Martyanov et al., 2023).

This study tackles a methodological rather than experimental gap by developing a reproducible Python prototype that integrates selected findings from the literature with exploratory regression modelling, visualisation and scenario analysis.

The project contributes a transparent, critical implementation of a simplified modelling workflow. It does not introduce a new diamond-growth process or optimisation algorithm.

2.9 Literature Review Summary

The literature shows that MPCVD diamond growth depends on interacting physical and chemical processes (Butler et al., 2009).

Methane concentration and substrate temperature are important variables, but their effects depend on the wider experimental conditions (Li et al., 2006; Martyanov et al., 2023).

Published experimental observations provide evidence of non-monotonic relationships between methane concentration and diamond growth rate (Martyanov et al., 2023).

The existing computational model serves as the basis for testing such nonlinear relationships; however, limited data and process variables prevent conclusive predictions or causal conclusions.

Chapter 3: Data and Mathematical Methodology

3.1 Introduction

This chapter outlines the experimental data and mathematical models used to explore the relationships between selected MPCVD parameters and diamond growth rate.

The method draws on selected literature findings and regression modelling performed in Python.

We consider two quantitative approaches: multiple linear regression and quadratic regression. We also estimate deposited diamond thickness from the predicted growth rate and deposition duration.

The analysis is exploratory because the dataset is small and does not fully represent the experimental conditions.

3.2 Experimental Data

The dataset comprises six selected growth-rate observations reported by Martyanov et al. (2023) for polycrystalline diamond films produced using microwave plasma.

The selected observations are:

Substrate temperature (°C)	Methane concentration (%)	Growth rate (µm/h)
800	4	0.60
800	20	3.41
800	100	0.60
900	3	2.80
900	40	4.10
900	100	2.10

Table 1. Selected MPCVD diamond-growth observations reproduced from Martyanov et al. (2023).

These values have been compared with the published article text and the local CSV.

The dataset is stored as mpcvd_experimental_data_v01.csv.

The variables used for regression are methane concentration, substrate temperature and diamond growth rate.

The original study also included reactor conditions not fully represented in the CSV, including pressure and microwave power (Martyanov et al., 2023).

Consequently, the dataset should not be interpreted as a controlled two-factor experiment in which only methane concentration and substrate temperature varied.

3.3 Data Preparation

The CSV dataset is imported into Python using pandas.

The numerical predictor variables are extracted from the dataset and organised into the input matrix.

The measured growth-rate values form the response vector.

The predictor and response variables used in each regression model are summarised below.

Multiple Linear Regression

- Inputs: Methane concentration (%) and substrate temperature (°C).
- Output: Diamond growth rate (µm/h).

Quadratic Regression

- Input: Methane concentration (%) and its squared term.
- Output: Diamond growth rate (µm/h).

The available dataset contains only six observations. This severely limits assessment of the stability and generalisability of the fitted relationships.

No claim is made that the selected observations adequately represent the full operating range of an MPCVD reactor.

3.4 Multiple Linear Regression

Multiple linear regression is used to investigate whether diamond growth rate can be represented as an additive linear function of methane concentration and substrate temperature.

The model is:

Predicted growth rate = $\beta_0 + \beta_1 \times$ methane concentration + $\beta_2 \times$ substrate temperature

Where:

- β_0 is the intercept.
- β_1 is the fitted methane coefficient.
- β_2 is the fitted temperature coefficient.

The regression coefficients are estimated from the six observations using ordinary least squares.

Ordinary least squares estimates the coefficients by minimising the sum of squared residuals between the observed and predicted growth rates.

Fitted coefficients show the statistical relationship in the selected dataset. They do not establish independent causal effects or physically universal growth laws.

3.5 Quadratic Regression

A separate quadratic regression model explores nonlinear behaviour between methane concentration and growth rate.

The model is:

Predicted growth rate = a + b × methane concentration + c × methane concentration²

Where a, b, and c are fitted coefficients.

The quadratic term allows the mathematical relationship to curve, potentially representing an increase followed by a decrease in growth rate.

This is motivated by published observations showing non-monotonic behaviour across methane concentrations (Martyanov et al., 2023).

However, the model excludes substrate temperature and other reactor conditions.

Its predictions are therefore exploratory and should not be treated as a physically validated description of MPCVD growth.

3.6 Estimation of Diamond Thickness

An illustrative thickness calculation is included in the computational system.

The calculation assumes a steady growth rate throughout the deposition process.

Estimated thickness = Predicted growth rate × Growth duration

Where:

- Thickness is measured in micrometres.
- Growth rate is measured in micrometres per hour.
- Duration is measured in hours.

For the previously recorded reference scenario:

- Methane concentration = 20%
- Substrate temperature = 800 °C
- Growth duration = 5 hours
- Predicted growth rate = 1.762 µm/h
- Estimated thickness = 1.762 × 5 = 8.810 µm

The thickness calculation is mathematically consistent, but it does not validate the predicted growth rate.

Consequently, while the illustrative thickness calculation provides a mathematically consistent estimate, its validity cannot be established without experimental confirmation. This limitation underscores a key conclusion of the study: computational predictions of film thickness based on simplified growth-rate models require empirical validation before they can support reliable scientific or engineering decisions.

3.7 Exploratory Scenario Analysis

The Python prototype also supports exploratory comparison of model outputs across selected input combinations.

This functionality lets users examine how fitted predictions change as methane concentration and substrate temperature vary.

Such comparisons help illustrate the regression model's mathematical behaviour.

However, the highest predicted growth rate within a computational search grid is not necessarily the experimentally optimal growth condition.

The results depend on the selected model, its training data and the investigated parameter range.

3.8 Model Evaluation and Reproducibility

The software implementation includes automated tests for selected software functions.

These tests help ensure the software behaves consistently with its intended calculations.

A reproducibility script and documented Python dependencies support repeating computational procedures.

In-sample evaluation calculates performance from predictions for the same observations used to fit the model. These measures describe how closely a model reproduces its training data, but they may give an optimistic impression of predictive performance.

Leave-one-out cross-validation (LOOCV) provided a more direct assessment of prediction for held-out observations. With six observations, each model was fitted six times: each fit used five observations for training and the remaining observation for testing. The six held-out prediction errors were aggregated using MAE, RMSE and R^2.

Unlike in-sample evaluation, each LOOCV prediction was generated by a model that was not fitted to the observation being predicted. LOOCV therefore provided limited evidence about generalisation while using the available data efficiently. However, estimates based on only six observations may be unstable because each observation strongly influences a model trained on the other five. LOOCV supplements, but does not replace, validation using an independent experimental dataset.

However, software testing does not establish scientific predictive accuracy.

A more detailed model evaluation would require additional independent observations, uncertainty analysis and evaluation of experimentally measured growth rates.

3.9 Methodological Restrictions

The main limitations of the procedure are:

1. The dataset contains only six selected observations.
2. The experimental conditions are not fully represented.
3. Pressure, microwave power and additional variables are omitted.
4. The observations may contain measurement uncertainty.
5. Regression coefficients cannot be interpreted as independent causal effects.
6. Quadratic regression may fit the small dataset without generalising to new observations.
7. Predicted growth rates may become physically unrealistic.
8. The constant-growth-rate thickness approximation is simplified.
9. The models have not undergone independent experimental testing.

These limitations are important when interpreting the computational results.

3.10 Chapter Summary

This chapter presented the data and mathematical methods used in the exploratory MPCVD modelling project.

We used multiple linear regression to represent growth rate as a function of methane concentration and substrate temperature.

Quadratic regression was used to investigate a nonlinear relationship with methane concentration.

We applied a constant-growth-rate approximation to estimate film thickness.

The methodology provides a reproducible computational system, but its small dataset and incomplete representation of reactor conditions limit reliable scientific prediction and validated process optimisation.

Chapter 4: Software Implementation and Computational Architecture

4.1 Introduction

This chapter describes the software architecture and computational execution of the exploratory MPCVD diamond-growth modelling framework.

The prototype is implemented in Python and integrates experimental data processing, mathematical regression, scientific visualisation, interactive scenario analysis, and automated testing.

The software architecture separates the mathematical modelling functions from the interactive user interface.

4.2 Development Environment

This project was built using Python in Visual Studio Code.

Python scripts built the model functions, and Jupyter notebooks supported data exploration and visualisation.

The principal software libraries include:

- pandas for importing and processing tabular data.
- NumPy for numerical operations.
- scikit-learn for regression modelling.
- Matplotlib for scientific visualisation.
- Streamlit for developing the interactive application.
- pytest for automated software testing.

A Python virtual environment was used for managing project dependencies.
The list of dependencies is stored in the file requirements.txt.

4.3 Software Architecture

The project has a modular structure which includes data, modelling, interface, testing, and documentation modules.

The experimental data is saved in a CSV file in the data directory.

The mathematical modelling algorithms are coded in src/model.py.

The interactive program is coded in src/app.py.

Notebooks for exploratory analysis, results, and tests are kept in different directories.

This structure separates the experimental data, computational functions and user interface.

4.4 Data Loading and Preparation

The data-loading component imports the CSV dataset using pandas.

The dataset contains six selected growth-rate observations from Martyanov et al. (2023).

The numerical variables are extracted for use in regression modelling.

The multiple linear regression model uses methane concentration and substrate temperature as predictors, with diamond growth rate as the response variable.

A separate quadratic model investigates methane concentration and growth rate.

The data-loading process provides a reproducible starting point for subsequent calculations.

4.5 Regression Model Implementation

The principal regression model is implemented using scikit-learn.

Multiple linear regression estimates the relationship between methane concentration, substrate temperature, and diamond growth rate.

The fitted model predicts growth rates for the supplied input values.

A separate quadratic regression approach investigates a curved relationship between methane concentration and growth rate.

The regression coefficients are estimated from the available observations rather than prescribed from physical theory.

Consequently, the models should be interpreted as exploratory statistical approximations.

4.6 Growth-Rate Prediction and Thickness Calculation

The computational method can generate growth-rate predictions and estimate diamond thickness.

The prediction function accepts selected process parameters and returns a model-generated growth rate.

The thickness calculation uses the relationship:

Estimated thickness = Predicted growth rate × Growth duration

This calculation assumes that the predicted growth rate remains constant throughout the deposition period.

For the previously recorded reference scenario, a growth rate of approximately 1.762 μm/h over five hours produces an estimated thickness of 8.810 μm.

This is an illustrative computational result, not an experimentally measured film thickness.

4.7 Interactive Streamlit Application

We developed a Streamlit application to provide an accessible interface for exploring the mathematical model.

The interface lets users enter selected process parameters and view the corresponding model outputs.

The application also incorporates visualisations and exploratory scenario comparisons.

This functionality enables users to investigate how the fitted mathematical relationship responds to changes in input values.

However, the application does not constitute a validated industrial process-control system.

Its outputs remain dependent on the limited dataset and the assumptions of the fitted regression models.

4.8 Scientific Visualisation

Scientific visualisation is an important component of the computational system, allowing graphical examination of experimental observations and regression outputs.

The project uses Python's Matplotlib library to generate scientific plots. These visualisations support exploratory data analysis, comparison of mathematical models and interpretation of predicted diamond growth behaviour.

Experimental Data Visualisation

The experimental observations are compared by methane concentration and substrate-temperature series in Table 2 and the accompanying analysis.

This representation allows examination of the empirical trends within each temperature series. However, differences between the series reflect not only changes in substrate temperature but also other reactor conditions, indicating that the variables are not independent (Martyanov et al., 2023).

Regression Model Comparison

Tables 3 and 4 compare the multiple linear and quadratic regression models using in-sample and LOOCV performance measures.

This comparison shows differences in model behaviour and the limitations of applying regression methods to the small experimental dataset.

Interpretation and Limitations

All visualisations are for exploratory interpretation, not for establishing causal relationships or independently validated predictions.

The visualisation scripts form part of the reproducible computational workflow; the present report interprets the recorded numerical outputs and tables.

4.9 Automated Testing and Verification

Automated testing was incorporated into the Python framework to assess the stability and dependability of its numerical functions. The tests were implemented using pytest and organised within the project's tests directory.

The testing process examined key components of the software, including dataset loading, regression predictions, diamond thickness calculations and input validation.

We executed 19 automated tests successfully, with no failures reported during the final verification run.

In addition to automated testing, we reproduced and evaluated both regression models' performance measures using the recorded results. We also performed a reference calculation using a methane concentration of 20%, a substrate temperature of 800 °C, and a deposition duration of five hours.

The implementation reproduced a predicted growth rate of 1.762 µm/h and an estimated diamond thickness of 8.810 µm.

These checks support the software's computational consistency. However, passing automated tests does not demonstrate that the regression models precisely predict diamond growth under new experimental conditions. Independent empirical confirmation would be required to establish their physical predictive reliability.

4.10 Software Limitations and Implementation Safeguards

Several limitations were observed while developing the computational system. These arise from both the restricted experimental dataset and the mathematical assumptions underlying the regression models.

The dataset contains only six selected experimental observations. Therefore, the fitted models depend on each data point and cannot be reliably generalised beyond the collected data.

Moreover, the dataset does not fully represent variations in reactor pressure, microwave power, and other experimental factors that can affect diamond growth (Martyanov et al., 2023). Therefore, differences between the two-temperature series cannot be attributed solely to substrate temperature.

The regression models may also produce physically unrealistic predictions, including negative growth rates, when applied to unsuitable input combinations. To address this, the software includes input validation and scientific warnings in the interface. This helps users understand the model's limited applicability but does not reduce the scientific uncertainty behind it.

Thus, the Streamlit application is intended for exploratory research and education, not for validated industrial optimisation.

Future development would require a larger experimental dataset, additional reactor parameters, more robust predictive models and independent validation by previously unseen experimental records.

Chapter 5: Results and Analysis

5.1 Introduction

This chapter presents the selected published MPCVD diamond-growth observations and examines the computational results generated by the exploratory regression framework.

The analysis examines relationships among methane concentration, substrate temperature and diamond growth rate.

It also considers illustrative model estimates, estimated diamond thickness and the limitations of interpreting results derived from a small dataset.

We reproduced the computational results using the final Python implementation. Both regression models were evaluated using mean absolute error (MAE), root mean squared error (RMSE) and the coefficient of determination (R^2). In-sample metrics describe fit to the same six observations used for training, while leave-one-out cross-validation (LOOCV) provides held-out predictions by fitting each model six times, with one observation excluded on each run.

5.2 Experimental Growth-Rate Observations

The dataset contains six selected observations reported by Martyanov et al. (2023).

Temperature (°C)	Methane (%)	Growth rate (µm/h)
800	4	0.60
800	20	3.41
800	100	0.60
900	3	2.80
900	40	4.10
900	100	2.10

Table 2. Experimental diamond growth rate by methane concentration and substrate temperature, reproduced from Martyanov et al. (2023).

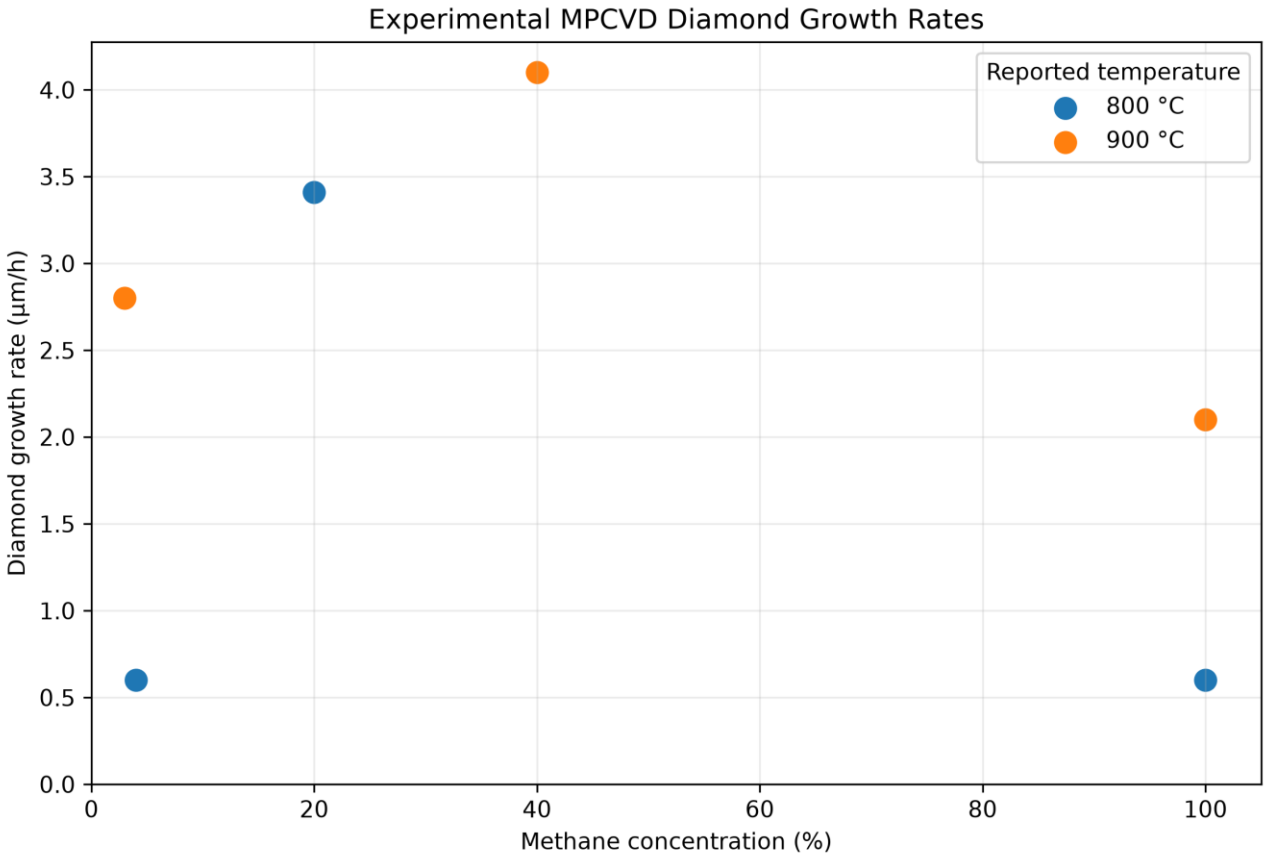


Figure 1. Experimental diamond growth rate (µm/h) as a function of methane concentration (%) for the 800 °C and 900 °C substrate-temperature series. Data reproduced from Martyanov et al. (2023). The series were obtained under different reactor conditions, so their differences must not be interpreted as the isolated effect of substrate temperature.

Within the selected 800 °C observations, the growth rate rises from 0.60 µm/h with 4% methane to 3.41 µm/h with 20% methane, then falls to 0.60 µm/h with 100% methane.

Within the selected 900 °C observations, the growth rate rises from 2.80 µm/h at 3% methane to 4.10 µm/h at 40% methane, then falls to 2.10 µm/h at 100% methane.

The data clearly show non-monotonic growth-rate behaviour with respect to the chosen methane concentrations.

However, they do not provide a complete relationship for all conditions.

5.3 Multiple Linear Regression Results

We fitted a multiple linear regression using methane concentration and substrate temperature as explanatory variables, with diamond growth rate as the response.

The model produced the following in-sample performance measures:

- Mean absolute error (MAE): 0.8890 µm/h
- Root mean square error (RMSE): 1.0164 µm/h
- Coefficient of determination (R^2): 0.4117

The R^2 value indicates that the fitted model accounts for approximately 41.17% of the variation in the six observed growth-rate values relative to a model that predicts the sample mean.

For the experiment conducted at 800 °C and 20% methane, the linear model predicts 1.762 µm/h, while the observed growth rate is 3.410 µm/h.

This discrepancy illustrates the linear model's limited ability to reproduce the selected experimental observations.

A simple additive linear model is therefore insufficient to represent the non-monotonic behaviour observed in the selected data.

5.4 Quadratic Regression Results

We fitted a quadratic regression model using methane concentration and its squared term as explanatory variables.

The model produced the following in-sample performance measures:

- Mean absolute error (MAE): 0.6743 µm/h
- Root mean squared error (RMSE): 0.7992 µm/h
- Coefficient of determination (R^2): 0.6363

The quadratic model attained a lower MAE and RMSE and a higher R^2 than the multiple linear regression model.

This indicates a closer fit to the six selected observations.

For 20% methane, the quadratic model predicted approximately 3.230 µm/h, compared with the reported observation of 3.410 µm/h at 800 °C (Martyanov et al., 2023).

However, the quadratic model uses methane concentration alone and does not independently account for substrate temperature.

The improved in-sample fit therefore does not establish better predictive performance for new MPCVD experiments.

Performance metric	Multiple linear regression	Quadratic regression
MAE (µm/h)	0.8890	0.6743
RMSE (µm/h)	1.0164	0.7992
R^2	0.4117	0.6363

Table 3. In-sample performance of the multiple linear and quadratic regression models.

The comparison shows that the quadratic model fits the six training observations more closely, but without independent validation, we cannot conclude its performance on new experimental data.

5.5 Leave-One-Out Cross-Validation Results

LOOCV was applied to both regression models using the six observations. Each model was fitted six times, with five observations used for training and the remaining observation used for testing in each iteration.

Metric	Linear LOOCV	Quadratic LOOCV
MAE (µm/h)	1.6663	1.3012
RMSE (µm/h)	1.8040	1.5232
R^2	−0.8534	−0.3212

Table 4. Leave-one-out cross-validation performance of the multiple linear and quadratic regression models.

The quadratic model achieved lower LOOCV MAE and RMSE and a less negative LOOCV R^2 than the linear model. However, both negative R^2 values indicate that the held-out predictions performed worse than predicting the mean of the observed growth rates. These results contrast with the more favourable in-sample metrics and reinforce the limited generalisability of both models.

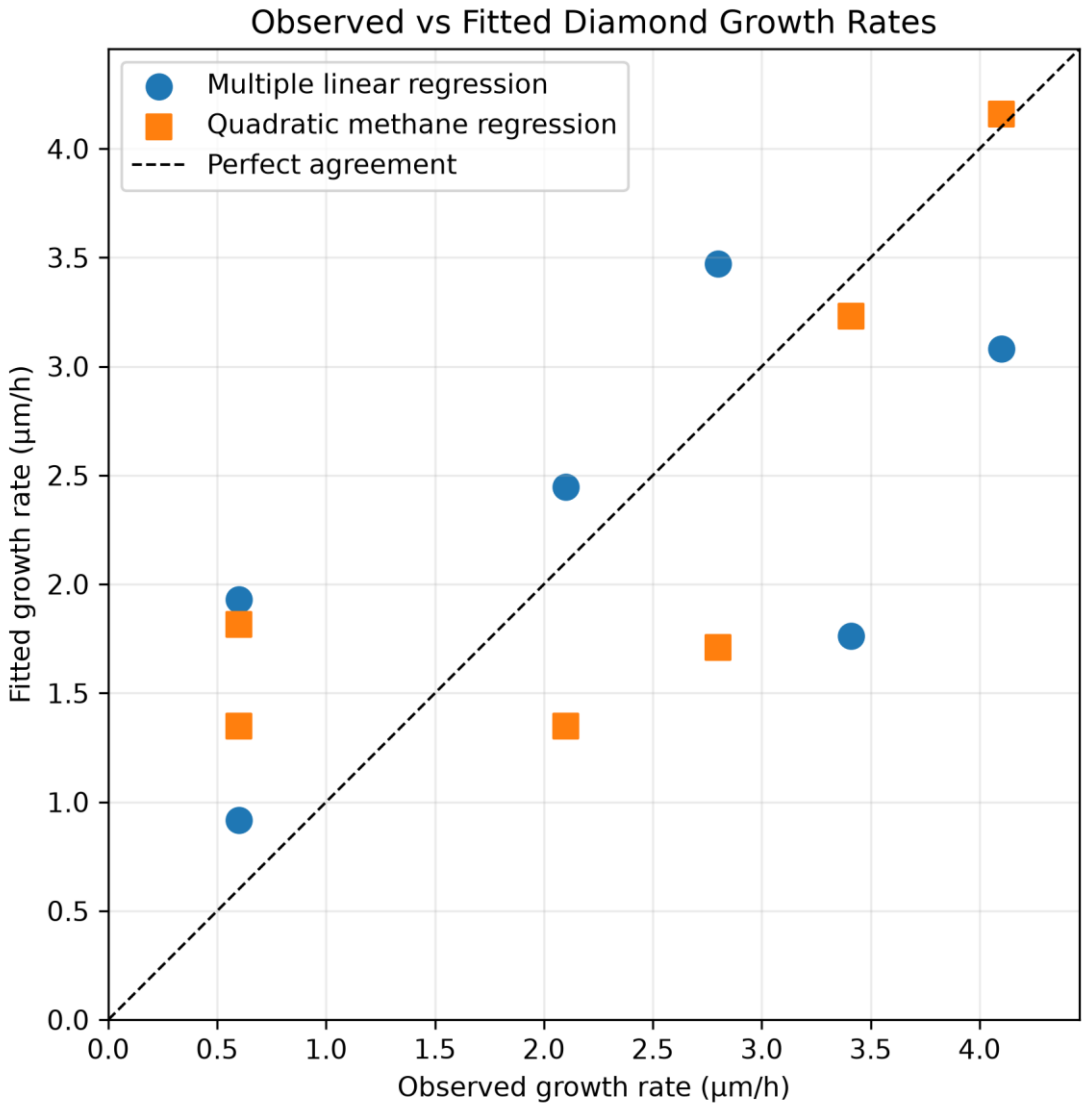


Figure 2. In-sample predictions from the multiple linear and quadratic regression models compared with six experimental MPCVD diamond-growth observations. This figure shows training-data fit, not independent predictive performance.

5.6 Illustrative Growth-Rate Prediction

A reference calculation was performed using the multiple linear regression model with the following input parameters:

- Methane concentration: 20%
- Substrate temperature: 800 °C
- Deposition duration: 5 hours

The multiple linear regression model predicted a growth rate of 1.762 µm/h.

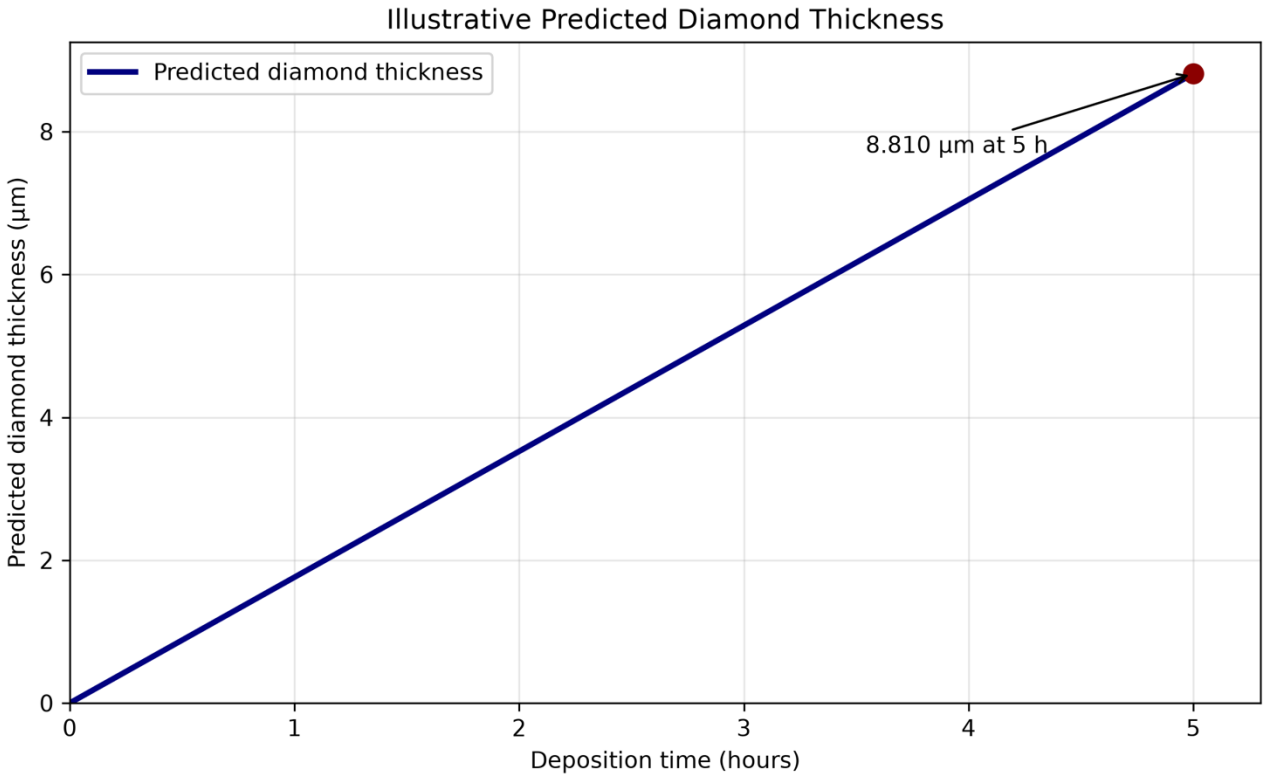


Figure 3. Model-estimated diamond thickness (µm) as a function of deposition time (h) at 20% methane and 800 °C. A constant predicted growth rate of 1.762 µm/h gives an estimated thickness of 8.810 µm after five hours; these are model outputs, not experimental measurements.

The experimentally observed growth rate at the same methane concentration and substrate temperature was 3.410 µm/h (Martyanov et al., 2023).

The 1.648 µm/h difference indicates a substantial fitting error in this experimental observation.

Although the software reproduced the recorded calculation, this does not establish the model's scientific validity. The value is a model prediction, not an experimentally validated growth rate.

5.7 Exploratory Scenario Comparisons

The Python application includes functionality to compare model predictions across selected input combinations.

These comparisons allow users to investigate how the fitted mathematical relationship responds to changes in methane concentration and substrate temperature.

However, the highest predicted growth rate within a computational scenario grid does not establish the physically optimal operating condition.

The result depends on the fitted model, the selected search range, and the underlying dataset.

5.8 Model Effectiveness Evaluation

As reported in Sections 5.3–5.5, the quadratic regression achieved lower MAE and RMSE and a higher R^2 than the multiple linear regression for both in-sample evaluation and LOOCV. However, the LOOCV errors were substantially larger than the in-sample errors, and both LOOCV R^2 values were negative. The quadratic model therefore performed better than the linear model in relative terms, but neither model demonstrated reliable held-out predictive performance.

The dataset contains only six observations, making the results vulnerable to any single observation and reducing the validity of the established dependencies.

The results distinguish exploratory in-sample fit from held-out LOOCV performance; neither constitutes independent experimental validation.

Future studies should include a more extensive set of experimental data and additional reactor parameters.

5.9 Interpretation of Results

The selected observations reported by Martyanov et al. (2023) show that the MPCVD diamond growth rate does not vary linearly with methane concentration.

They also show that growth behaviour differs between the two reported temperature series.

Differences between the two-temperature series cannot be attributed solely to substrate temperature because other experimental conditions also changed (Martyanov et al., 2023).

The computational prototype shows that mathematical models can be fitted, predictions generated, and visualisations developed.

It does not establish reliable physical prediction or validated process optimisation.

5.10 Chapter Summary

This chapter presented the selected published growth-rate observations and the exploratory modelling framework used to analyse them.

The selected observations show nonlinear growth behaviour, and the Python framework can fit, compare and visualise exploratory regression models. The quadratic model performed better than the linear model in both in-sample and LOOCV comparisons.

However, LOOCV produced substantially larger errors than the in-sample evaluation. The linear model returned MAE 1.6663 µm/h, RMSE 1.8040 µm/h and R^2 −0.8534; the quadratic model returned MAE 1.3012 µm/h, RMSE 1.5232 µm/h and R^2 −0.3212. Both negative R^2 values indicate poor held-out predictive performance.

The limited dataset, incomplete reactor information and absence of independent experimental validation therefore prevent reliable prediction; the next chapter discusses these limitations in detail.

Chapter 6: Discussion and Critical Assessment

6.1 Introduction

This chapter critically assesses the scientific and computing limitations of the exploratory modelling project on MPCVD diamond growth.

These include data quality, mathematical assumptions, experimental conditions, model accuracy, software reproducibility, and the need for future validation.

The project delivers a functioning computational framework, but its scientific contribution must be interpreted within four connected constraints: a six-observation dataset, incomplete reactor conditions, simplified regression assumptions and no independent experimental validation. Together, these constraints substantially restrict the reliability and generalisability of the results.

6.2 Evaluation of the Experimental Dataset

The dataset includes six growth rates selected from Martyanov et al. (2023).

The observations are traceable to the cited experimental study, but they form a small, selectively extracted dataset rather than a representative sample of MPCVD operating conditions.

Nevertheless, six observations are not enough for building a general predictive model.

Although regression coefficients can be estimated from six observations, additional data are needed to establish their stability and reliability.

Moreover, the chosen observations do not reflect the entire set of experimental results.

A larger and more representative dataset is therefore required for predictive modelling.

6.3 Limitations of the Experimental Conditions

The current model uses methane concentration and substrate temperature as explanatory variables.

However, MPCVD diamond growth also depends on reactor pressure, microwave power, gas composition, and substrate characteristics (Butler et al., 2009; Li et al., 2006).

The experimental study adjusted reactor conditions across temperature series (Martyanov et al., 2023).

The selected observations do not constitute a controlled two-factor experiment in which only methane concentration and substrate temperature varied.

In this respect, the physical meaning of the regression coefficients becomes limited.

Consequently, the fitted coefficients may reflect omitted reactor variables and cannot isolate the independent effect of methane concentration or substrate temperature.

6.4 Evaluation of Multiple Linear Regression

Multiple linear regression provides an intuitive baseline model.

It is straightforward to implement and computationally inexpensive.

However, the model presumes an additive linear relationship between methane concentration, substrate temperature and growth rate.

This assumption may be inappropriate for MPCVD growth, which is governed by nonlinear plasma chemistry and surface reactions (Butler et al., 2009).

The selected published observations also exhibit non-monotonic behaviour across methane concentrations (Martyanov et al., 2023).

Therefore, multiple linear regression should be regarded as a baseline exploratory method rather than a complete physical description of diamond growth.

6.5 Evaluation of Quadratic Regression

Quadratic regression provides a more flexible description of the relationship between methane concentration and growth rate.

Its curved form can represent an initial rise followed by a decline.

This method is relevant because the experimental data illustrate a non-monotonic pattern in the growth rate (Martyanov et al., 2023).

However, the quadratic model uses methane concentration alone, excluding substrate temperature and other reactor parameters. Its fitted relationship may therefore combine observations from different experimental regimes.

The model may also produce unrealistic predictions outside the observed methane range, so its improved in-sample fit should not be interpreted as physical validity.

A visually convincing fitted curve does not establish scientific validity (Motulsky and Ransnas, 1987, pp. 365–374).

6.6 Model Reliability and Overfitting

Overfitting occurs when a model learns patterns specific to the available dataset that do not generalise to new data points (James et al., 2021).

With only six observations, we cannot reliably assess the stability and generalisability of the fitted relationships, particularly for the more flexible quadratic model.

A model can fit the sample dataset well while failing to perform well on new, independent data (James et al., 2021).

Model evaluation must distinguish in-sample fit from predictive performance on independent data (Pratiwi et al., 2024).

The LOOCV results provide direct evidence of this limitation. The multiple linear regression produced a cross-validated R^2 of −0.8534, while the quadratic regression produced a cross-validated R^2 of −0.3212. A negative cross-validated R^2 means that the held-out predictions were less accurate, in terms of squared error, than a simple benchmark that predicts the mean observed growth rate. Although the quadratic model performed better than the linear model in relative terms, neither model demonstrated reliable predictive performance for excluded observations.

The difference between the favourable in-sample R^2 values and the negative LOOCV R^2 values indicates model instability. Each LOOCV fit uses only five observations, so removing one record can substantially alter the fitted coefficients and predictions. This sensitivity is especially important because the dataset combines two-temperature series obtained under differing reactor conditions and does not include all influential variables.

Overfitting risk is greatest for the quadratic model because its additional squared term increases flexibility relative to the available sample size. The model can follow curvature within the six training observations, producing improved in-sample MAE, RMSE and R^2, without capturing a relationship that generalises to a held-out observation. The linear model is less flexible but remains unstable because it estimates an intercept and two coefficients from only six records. Consequently, neither the closer quadratic fit nor the simpler linear form should be interpreted as evidence of a dependable physical growth law.

LOOCV strengthens the critical assessment by exposing the gap between training-data fit and held-out prediction, but it does not resolve the underlying evidence limitations. Reliable model selection requires a larger

and more representative dataset, complete reactor-condition variables, uncertainty analysis and independent experimental validation.

Independent experimental data are still required to evaluate predictive performance under genuinely unseen MPCVD conditions.

6.7 Physical Plausibility and Extrapolation

The regression models are mathematical approximations and do not explicitly represent the full physical mechanisms of MPCVD (Butler et al., 2009).

Consequently, they may produce unrealistic outputs for certain input combinations.

Potential problems include negative predicted growth rates, excessive predictions and extrapolation beyond the experimental conditions.

Future software development should include clearer warnings for unsupported inputs and checks for physical plausibility.

However, computational restrictions alone cannot replace experimentally validated physical modelling.

6.8 Limitations of Thickness Estimation

The prototype estimates diamond thickness by multiplying the predicted growth rate by the deposition duration.

This calculation assumes a constant growth rate throughout deposition.

Actual growth behaviour may vary over time because of nucleation, changing surface conditions and other process effects (Butler et al., 2009).

The thickness calculation is therefore a suitable illustrative estimate rather than a validated prediction of final film thickness.

Experimental thickness measurements would be required to assess its accuracy.

6.9 Software Testing and Reproducibility

Reproducibility depends on preserving the computational chain so that the same data, code, dependencies and test procedures can be rerun (Beaulieu-Jones and Greene, 2017, pp. 342–346).

The computational workflow relies on documented datasets, Python scripts, specified dependencies and automated testing. Together, these elements support software transparency, reproducibility and reliability.

Automated tests were implemented using pytest.

All **19 tests passed**, including those for loading data, regression predictions, diamond thickness calculations, exploratory scenarios, and input validation.

The additional tests confirmed that the multiple linear and quadratic regressions replicated the expected performance measures. The reference calculation also replicated a predicted growth rate of 1.762 µm/h and a thickness of 8.810 µm.

These results support consistency between the tested software functions and their intended mathematical calculations.

Nevertheless, computational verification must be distinguished from scientific validation. Passing the tests shows that the implemented functions reproduce their specified calculations; it does not show that the regression models accurately reproduce MPCVD diamond growth.

Thus, the framework provides reproducible exploratory computation but not experimental validation of prediction.

6.10 Evaluation of the Interactive Application

The Streamlit interface provides a good platform for exploring the mathematics behind the model fits. It lets users explore process parameters, make predictions, and visualise results. This is a good example of how computational methods can help in exploratory engineering research.

However, the tool does not provide validated recommendations for industrial operations. Its outputs are limited to the dataset used for prediction. This means that the interface should highlight the exploratory nature of its predictions.

6.11 Future Improvements

The principal priorities are to expand the experimental dataset, include additional reactor variables, evaluate uncertainty and independent predictive performance, strengthen input-domain safeguards, and consider physical and multi-objective modelling. These recommendations are developed in detail in Section 7.5.

6.12 Chapter Summary

The project demonstrates a functional exploratory computational approach to modelling MPCVD diamond growth.

The advantages of the model are modular software architecture, reproducible computations, visualisation, and documentation.

Nevertheless, the small dataset, incomplete experimental conditions, simplified regression assumptions and lack of independent validation limit the scientific validity of the results. The negative LOOCV R^2 values show that both models generalise poorly to held-out observations, while the gap between in-sample and cross-validated performance indicates instability and overfitting risk.

Thus, the developed models should be treated as unstable exploratory approximations and the software as a research prototype, not an engineering optimisation tool.

These limitations also define a clear route for extending the prototype through broader experimental data, additional reactor variables, uncertainty analysis and independent validation.

Chapter 7: Conclusion and Future Research

7.1 Research Summary

This research applied mathematical modelling and Python-based computational methods to the analysis of MPCVD diamond growth.

The primary objective was to develop an exploratory computational framework to analyse experimental growth data, compare regression models, and estimate diamond growth rates under selected process conditions.

Six experimental observations reported by Martyanov et al. (2023) were used to fit and evaluate two regression models: multiple linear regression and quadratic regression.

We implemented both models in Python and incorporated them into an interactive Streamlit web application, allowing users to explore how predicted diamond growth changes with selected process inputs.

To estimate model performance, we calculated the mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). In addition, we performed automated software testing and reproducibility checks.

The developed system is a practical example of combining mathematical modelling, data analysis, and software development to solve a materials engineering problem.

Nevertheless, the small dataset and incomplete information about experimental conditions limit the reliability and applicability of the results.

7.2 Key Findings

The computational analysis produced several findings concerning regression model performance, experimental diamond growth behaviour and the limitations of data-driven prediction.

1. Comparison of Regression Models

We fitted two regression models to the six experimental observations. As shown in Tables 3 and 4 (Sections 5.4 and 5.5), quadratic regression achieved lower MAE and RMSE and a higher R^2 than multiple linear regression in both the in-sample and LOOCV evaluations.

The LOOCV results were substantially weaker than the in-sample results: the linear model produced an MAE of 1.6663 μm/h, RMSE of 1.8040 μm/h and R^2 of −0.8534, while the quadratic model produced an MAE of 1.3012 μm/h, RMSE of 1.5232 μm/h and R^2 of −0.3212. Both negative R^2 values indicate poor held-out predictive performance.

2. Experimental Growth Behaviour

The experimental observations showed that diamond growth rate varied with methane concentration. The observed relationships were not uniformly linear across the selected conditions (Martyanov et al., 2023).

The two-temperature series also differed under other reactor conditions, so their differences cannot be interpreted as the isolated effect of substrate temperature (Martyanov et al., 2023).

3. Illustrative Prediction

Based on the multiple linear regression model, for a methane concentration of 20% and substrate temperature of 800 °C, the growth rate was 1.762 μm/h.

Assuming a constant predicted growth rate over five hours, the illustrative estimated diamond thickness is 8.810 μm.

However, the experimentally observed growth rate at the same methane concentration and substrate temperature was 3.410 μm/h, showing a substantial difference from the model prediction.

4. Computational Verification

The Python code correctly reproduced the previously recorded performance metrics of the regression models and the example prediction.

All 19 software tests were completed successfully, confirming that the tested functions performed as intended. This verification of computational correctness does not constitute scientific validation of the regression models or their predictive accuracy.

5. Overall Finding

The project demonstrates the feasibility of a reproducible, interactive exploratory computational prototype for MPCVD diamond-growth analysis.

Its principal achievement is integrating experimental data, mathematical modelling, cross-validation, visualisation and software testing; it is not a validated industrial optimisation system.

Overall, although the quadratic model performed better than the multiple linear regression in relative terms, the negative LOOCV R^2 values, small dataset and incomplete reactor information prevent either model from supporting reliable industrial prediction or optimisation.

7.3 Research Contributions

This project has made several contributions to the exploratory computational analysis of MPCVD diamond growth.

1. Development of Computational Framework

We developed a computational framework to analyse the selected experimental data and provide mathematical predictions. The project integrated data processing, regression modelling, and diamond growth thickness calculations into a well-structured software framework.

2. Comparison of Mathematical Models

The study implemented and compared multiple linear and quadratic regressions using the same experimental dataset. This comparison shows how model form affects in-sample fit and highlights the importance of considering nonlinear relationships in exploratory growth modelling.

3. Interactive Scientific Software

We created an interactive software application using Streamlit. It allows users to enter selected process parameters and evaluate the model's predictions for diamond growth rate and thickness.

It provides a convenient way to explore the mathematical model without working directly with the underlying Python scripts.

4. Reproducibility and Software Verification

The current project uses documented experimental data, reusable Python scripts, versioning, and automated tests.

The successful completion of 19 automated tests supports the repeatability and internal consistency of the implemented computational framework.

5. Identification of Research Limitations

The results show the difficulty of building predictive models from small experimental datasets, especially since some of the reactor's key parameters were not captured in the experiment.

Separating computational verification from experimental validation is an important step toward building a basis for future studies.

Overall Contribution

The main contribution of the work is a reproducible proof-of-concept computational approach involving mathematical modelling, experimental data processing, and software development applied to materials engineering.

The research has not produced a new physical theory of diamond growth or process optimisation. It provides a basis that can be further expanded.

7.4 Research Limitations

The principal limitations are the six-observation dataset, incomplete reactor-condition variables, simplified statistical models, the absence of independent experimental validation, restricted extrapolation and the lack of multi-objective optimisation.

1. Limited Experimental Dataset

The research used six experimental observations reported by Martyanov et al. (2023). The small number of observations limits the models' ability to capture more complex dependencies and makes them sensitive to specific data points.

2. Incomplete Representation of Reactor Conditions

The dataset includes methane concentration, substrate temperature, and diamond growth rate. Still, it does not account for other factors that affect diamond growth, such as reactor pressure and microwave power.

In addition, two different experimental temperature datasets were provided under different reactor conditions (Martyanov et al., 2023). Consequently, the observed differences may also reflect changes in reactor conditions other than substrate temperature.

3. Simplified Mathematical Models

Both multiple linear regression and quadratic regression are simple mathematical representations of the experimental data. They do not represent the process physics, such as plasma chemistry or surface reactions (Butler et al., 2009).

4. Absence of Independent Model Validation

Both regression models were assessed using in-sample metrics and leave-one-out cross-validation on the same six-observation dataset.

The in-sample metrics describe training-data fit, while LOOCV provides held-out predictions; neither constitutes validation against an independent experimental dataset.

No independent experimental dataset was available to validate the models under genuinely unseen MPCVD conditions.

5. Restricted Predictive Applicability

The regressions can predict unphysical values when applied beyond the suitable range of inputs.

Although the paper includes software validation and scientific warnings, these do not ensure the physical feasibility of the predicted values.

6. Limited Optimisation Capability

The framework allows exploratory analysis of how some inputs affect the growth results.

However, it does not validate a multi-objective optimisation approach or comprehensively evaluate diamond quality, energy efficiency, costs, and reactor capacity.

It should therefore be considered a proof-of-concept modelling framework rather than an optimisation system for industrial use.

Overall Assessment

These limitations preserve the project's value as a reproducible methodological demonstration while restricting its conclusions to exploratory model behaviour rather than reliable engineering prediction.

7.5 Recommendations for Future Research

The current study's results and limitations suggest several avenues for future research and development.

1. Expanding the Experimental Dataset

Future work must include a much larger experimental dataset with observations gathered across a wide range of MPCVD settings.

Whenever possible, future datasets should include replicate experiments and fully described reactor setups, enabling more reliable statistical analysis and evaluation of model performance on independent observations.

2. Additional Influential Variables

The modelling framework could be further developed by accounting for other important reactor variables, such as reactor chamber pressure, microwave power, the hydrogen-to-methane ratio in the gas, substrate temperature, and deposition time.

3. Developing New Models for Prediction

The current work may be expanded into comparative studies of current models and other regression analysis methods, such as regularised regression, Gaussian process regression, and other machine-learning approaches.

Such approaches may be more flexible in modelling nonlinear relationships, but this must be tested with larger datasets.

4. Independent Experimental Validation

The next essential step is to validate the model predictions against independent experimental data not used for model fitting. This would test whether the framework generalises beyond its training observations and help define the conditions under which its predictions are valid.

5. Multi-Objective Process Optimisation

The framework can be developed so that it can study optimisation goals beyond growth rate.

Possible goals include diamond quality, deposition time, energy usage, and production cost.

Multi-objective optimisation will allow us to study trade-offs between these conflicting goals while accounting for the reactor's physical limitations. This development will require additional experiments and their analysis.

6. Physical Modelling Integration

Another interesting research path would be to integrate the proposed statistical modelling with well-known physical models of plasma chemistry, gas-phase reactions and diamond surface growth (Butler et al., 2009).

7. Future Software Development

Future software development should include improved visualisations, model uncertainty estimates, stronger input validation and clearer warnings about unsupported predictions.

This will allow more transparent, well-informed use of the computational tool.

Overall Research Objective

The ultimate aim is to develop a scientifically validated decision-support tool that enables the study of MPCVD process conditions and possible trade-offs in growth rate, quality, and efficiency.

This will require further experimentation, mathematical modelling, and validation.

7.6 Final Conclusion

This investigation examined how mathematical modelling and computational methods can support the study of MPCVD diamond growth.

The project developed a Python framework for experimental-data analysis, regression modelling, growth-rate and thickness calculations, scientific visualisation and interactive exploration.

Using six selected experimental observations from Martyanov et al. (2023), the quadratic regression performed better than the multiple linear regression in both in-sample and LOOCV comparisons. However, both models produced negative LOOCV R^2 values and neither underwent independent experimental validation.

The Streamlit application, automated tests and reproducible procedures demonstrate the practical integration of mathematics, programming and materials engineering. The system supports exploratory analysis, hypothesis generation and the identification of future experiments.

Its reliability and generalisability remain limited by the small sample, incomplete reactor-condition data and simplified modelling assumptions.

Further experimental data, additional process variables, uncertainty analysis and independent validation are required before the framework can support engineering prediction, process control or optimisation.

In conclusion, the project establishes a reproducible exploratory foundation for analysing MPCVD diamond growth. Its contribution is methodological: it demonstrates an integrated computational workflow and defines priorities for future research rather than providing validated industrial recommendations.

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